Grade 8 | Name______________________
Unit 5 | Date______________________
Lesson 1 Practice | Period_____________________

Use the table of possible solutions for questions 1–2.

Possible Solutions				
-11.5	36	$6\frac{3}{8}$	-8	-0.1212
-25	-9.5	-7.75	$6\frac{2}{7}$	-17.3

Determine the values from the table that are solutions to the inequalities.

1. $-3 + \frac{x}{2} < -7$

   $+3 \qquad +3$ $\qquad$ $x < -8;\ -11.5,\ -25,\ -9.5,\ -17.3$

   $2 \cdot \frac{x}{2} < -4 \cdot 2$

2. $-3 + \frac{x}{2} > -7$

   $+3 \qquad +3$ $\qquad$ $x > -8;\ 36,\ 6\frac{3}{8},\ -7.75,\ 6\frac{2}{7},\ -0.1212$

   $2 \cdot \frac{x}{2} > -4 \cdot 2$

3. Rewrite $-3 + \frac{x}{2} < -7$ and $-3 + \frac{x}{2} > -7$ so that -8 is a solution. Explain your reasoning.
   Change $<$ and $>$ to $\leq$ and $\geq$ because greater than/less than <u>or equal to</u> means -8 will be included.

4. Inequalities have ______infinite______ solutions that are represented on the number line.

5. Tanner and and Jonah were solving the inequality $-3 - \frac{y}{7} < -5$.
   Jonah said the solution is $y < -14$, while Tanner said the solution is $y > -14$. Who do you agree with or do you agree with neither? Justify your reasoning mathematically.

   $-3 - \frac{y}{7} < -5$ $\qquad$ $-7 \cdot \frac{-y}{7} < -2 \cdot -7$ $\qquad$ $y > 14$

   $+3 \qquad +3$

   Neither because $-2 \cdot -7 = +14$

6. When multiplying or dividing by negative coefficient in an inequality, the resulting inequality symbol <u>will change to its opposite</u>.

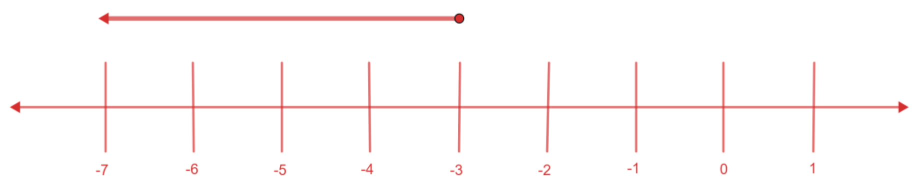
For questions 7–10, solve each inequality and graph the solution set.

7. $-8 \leq \frac{a}{-3} - 9$

$+9 \qquad +9$

$-3 \cdot 1 \leq \frac{a}{-3} \cdot -3$

$-3 \geq a$ or $a \leq -3$

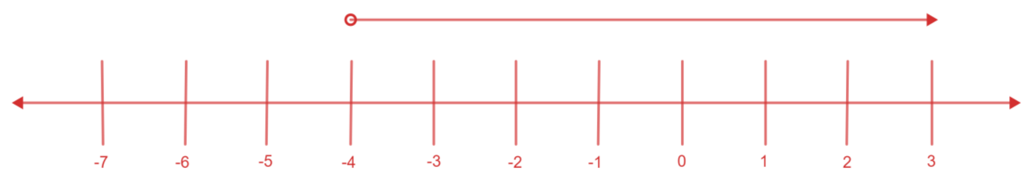


8. $9m + 16 > -20$

$-16 \qquad -16$

$\frac{9m}{9} > \frac{-36}{9}$

$m > -4$

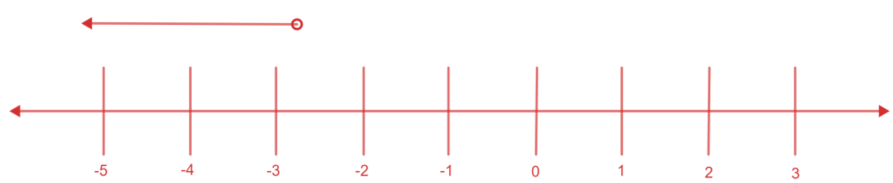


9. $2.8 < -1.3\text{p} - 0.5$

$+0.5 \qquad +0.5$

$\frac{3.3}{-1.3} < \frac{-1.3p}{-1.3}$

$-2\frac{7}{13} > p$ or $p < -2\frac{7}{13}$



10. $-2 + \frac{5}{4}x \geq -12$

$+2 \qquad +2$

$\frac{4}{5} \cdot \frac{5}{4}x \geq -10 \cdot \frac{4}{5}$

$x \geq -8$